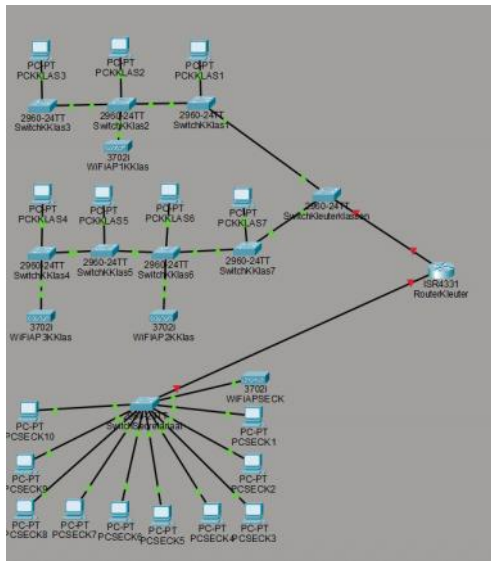
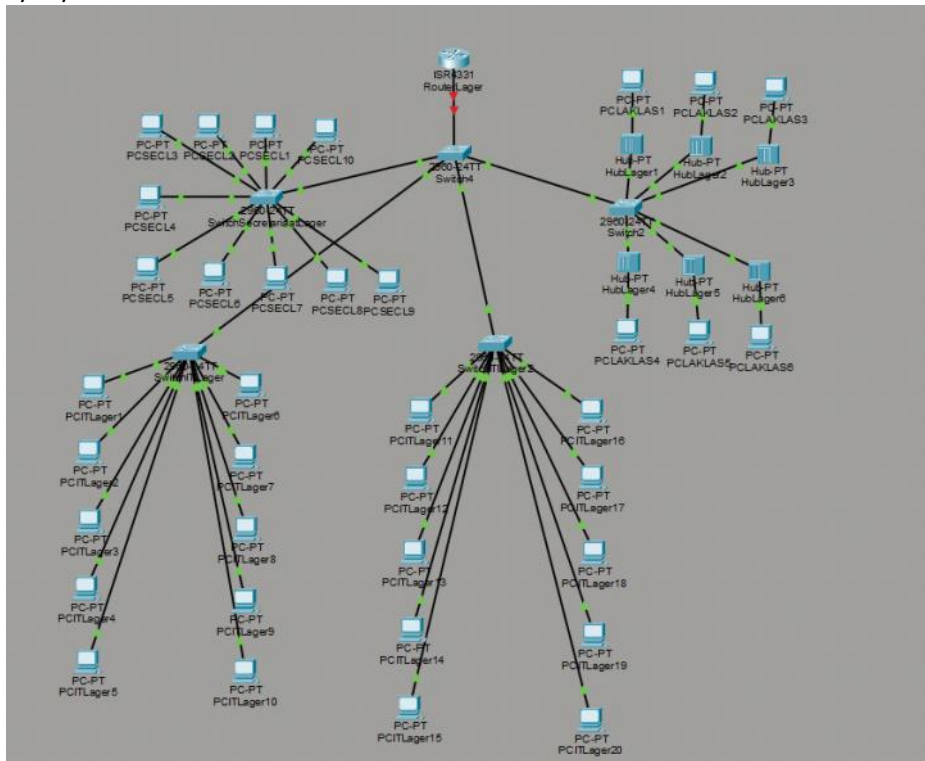


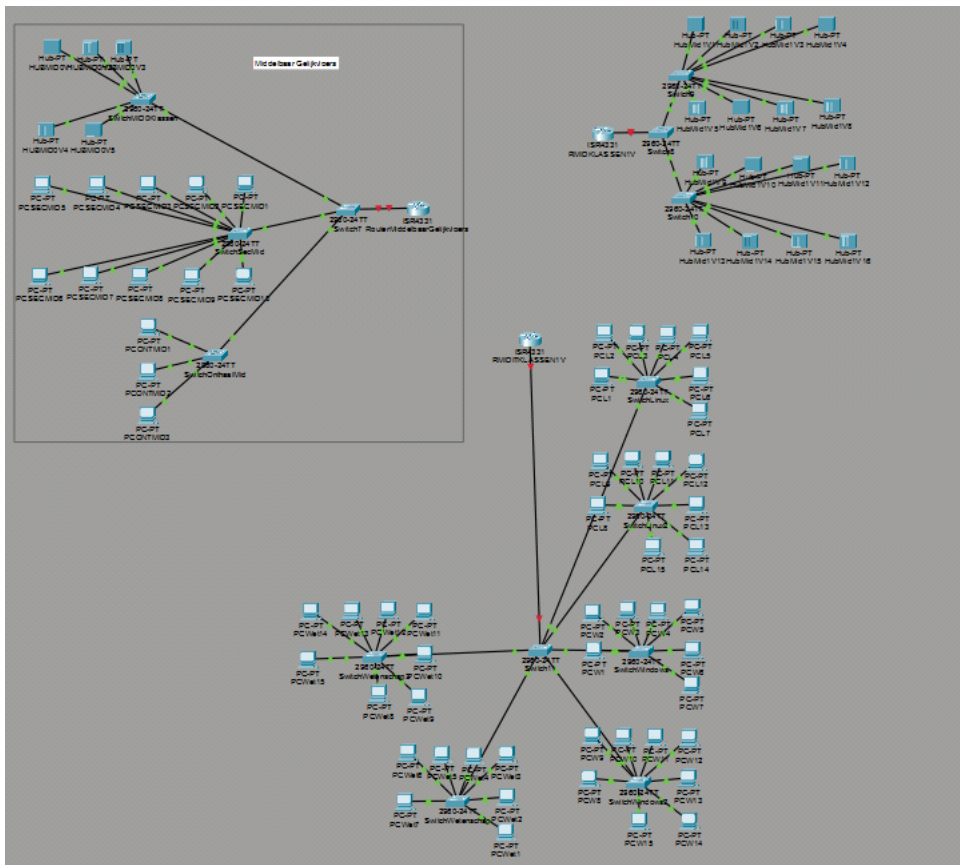
17 March 2026 11:33

25/02/2026:

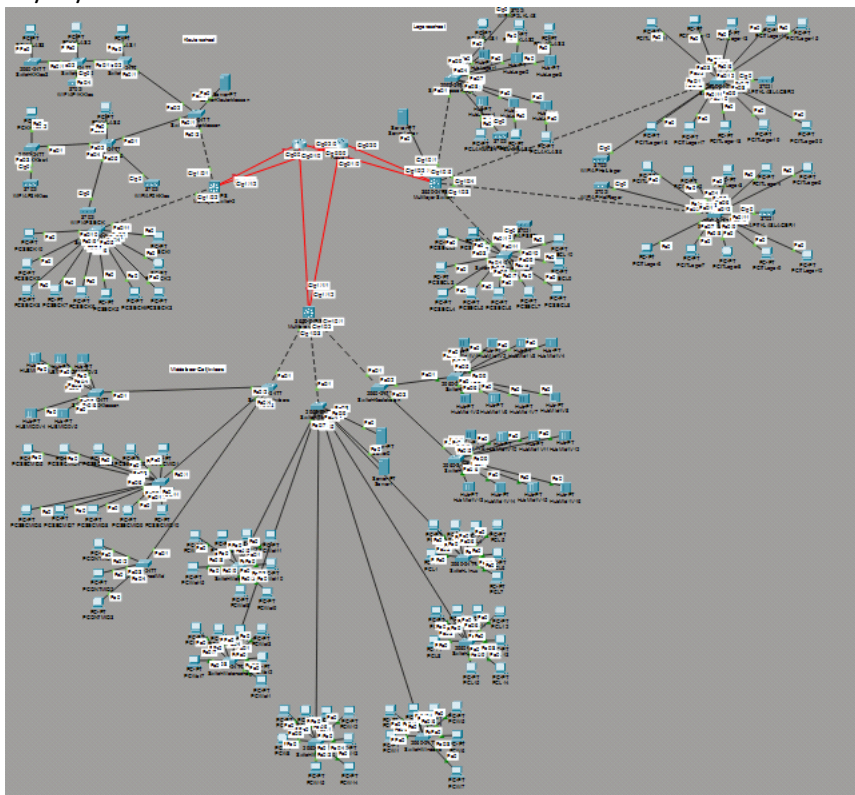


3/03/2026:

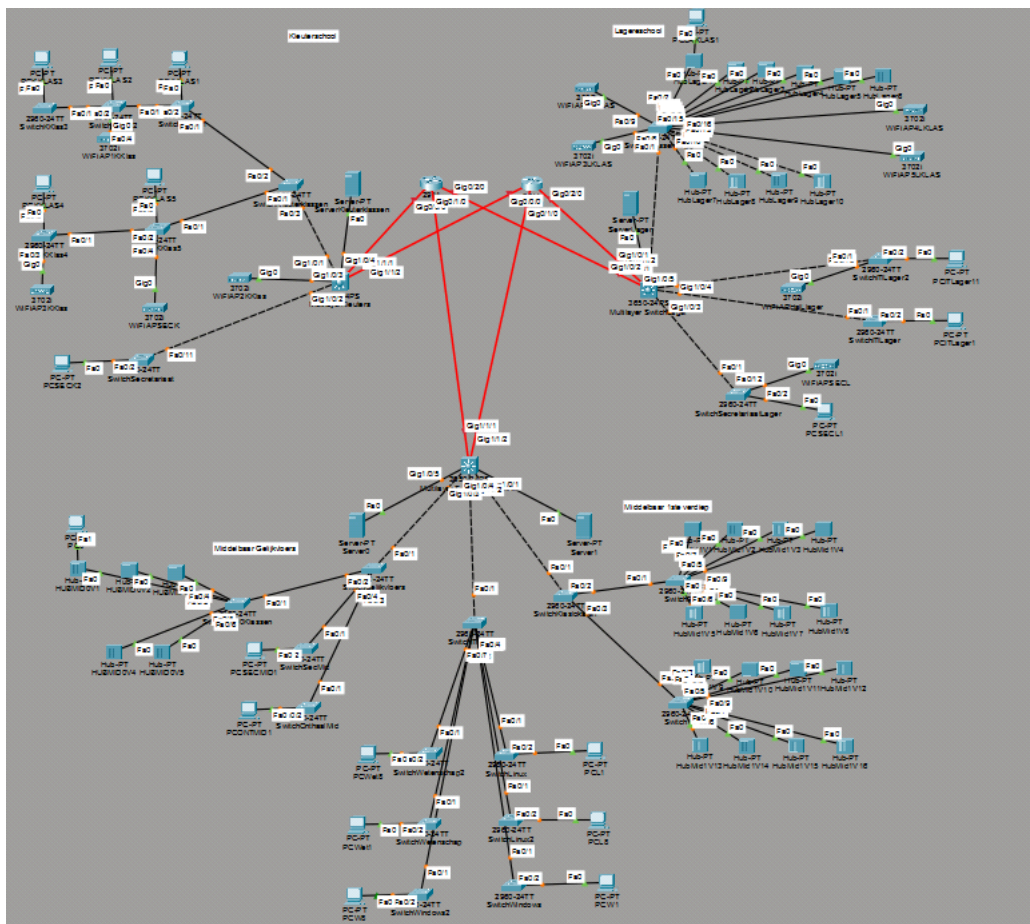




04/03/2026



Samenvoeging tot 1 groot netwerk, met 2 fiber ISP's ter redundancy.



File opgeschoont en verduidelijkt

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable
% Incomplete command.
Switch(config)#conf t
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)# name SecretariaatKleuters
Switch(config-vlan)#vlan 11
Switch(config-vlan)# name SecretariaatMiddelbaar
Switch(config-vlan)#vlan 12
Switch(config-vlan)# name SecretariaatLager
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name Klassen1verdiepMiddelbaar
Switch(config-vlan)#vlan 21
Switch(config-vlan)# name KlassenGelijkMiddelbaar
Switch(config-vlan)#vlan 22
Switch(config-vlan)# name KlassenLager
Switch(config-vlan)#vlan 23
Switch(config-vlan)# name KlassenKleuter
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name Servers
Switch(config-vlan)#vlan 40
Switch(config-vlan)# name WiFiKlassen31
Switch(config-vlan)#vlan 41
Switch(config-vlan)# name WiFiKlassen29
Switch(config-vlan)#vlan 42
Switch(config-vlan)# name WiFiKlassen30en29
Switch(config-vlan)#vlan 43
Switch(config-vlan)# name WiFiITLokalen
Switch(config-vlan)#vlan 44
Switch(config-vlan)# name WiFiKlassenGelijkMiddel
Switch(config-vlan)#vlan 99
Switch(config-vlan)# name Management
Switch(config-vlan)#exit
Switch(config)#sh
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

```

MLSkleuters>en
MLSkleuters>enable
MLSkleuters#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLSkleuters(config)#int gi1/0/2
MLSkleuters(config-if)#switchport access
% Incomplete command.
MLSkleuters(config-if)#switchport mode access
MLSkleuters(config-if)#switchport access vlan 10
MLSkleuters(config-if)#exit
MLSkleuters(config)#int gi1/0/1
MLSkleuters(config-if)#swi
MLSkleuters(config-if)#switchport mod
MLSkleuters(config-if)#switchport mode acc
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/2
(10), with SSecretariaat FastEthernet0/11 (1).

MLSkleuters(config-if)#switchport mode access
MLSkleuters(config-if)#sw
MLSkleuters(config-if)#switchport acc
MLSkleuters(config-if)#switchport access vlan 23
MLSkleuters(config-if)#exit
MLSkleuters(config)#int gi1/0/4
MLSkleuters(config-if)#swi
MLSkleuters(config-if)#switchport mo
MLSkleuters(config-if)#switchport mode ac
MLSkleuters(config-if)#switchport mode access
MLSkleuters(config-if)#sw
MLSkleuters(config-if)#switchport ac
MLSkleuters(config-if)#switchport access vlan 30
MLSkleuters(config-if)#exit
MLSkleuters(config)#int gi1/0/3
MLSkleuters(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1
(23), with SwitchKleuter FastEthernet0/3 (1).
sw
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/2
(10), with SSecretariaat FastEthernet0/11 (1).
i
MLSkleuters(config-if)#switchport mo
MLSkleuters(config-if)#switchport mode ac
MLSkleuters(config-if)#switchport mode access
MLSkleuters(config-if)#s
MLSkleuters(config-if)#sw
MLSkleuters(config-if)#switchport acc
MLSkleuters(config-if)#switchport access vlan 23
MLSkleuters(config-if)#

```

VLAN acces toewijzen op de MLS switch naar de kleuterklas. Zo krijgt iedere distribution switch de jusite VLAN binnen. Dit deed ik ook voor de switches van de Lagere en Middelbare scholen

```

MLSkleuters(config)#! VLAN 10
MLSkleuters(config)#int vlan 10
MLSkleuters(config-if)#ip add 10.2.10.2 255.255.255.224
MLSkleuters(config-if)#standby 10 ip 10.2.10.1
MLSkleuters(config-if)#standby 10 priority 110
MLSkleuters(config-if)#standby 10 preempt
MLSkleuters(config-if)#no shut
MLSkleuters(config-if)#

```

Per multilayer switch de VLAN's een IP geven via SVI zo dienen de switches als default gateway, daarnaast al priority ingesteld om later naar meer redundancy toe te werken.

```

MLSkleuters#enable
MLSkleuters#conf t
Enter configuration commands, one per line. End with CNTL/Z.
MLSkleuters(config)#ip route 0.0.0.0 0.0.0.0 10.2.250.254
MLSkleuters(config)#exit
MLSkleuters#

```

Deze config stuurt alle non-lokale pakketten naar het virutele IP dat gedeeld wordt door beide routers.


```

MLSkleuters(config)#int gil/1/2
MLSkleuters(config-if)#no swi
MLSkleuters(config-if)#no switchport
MLSkleuters(config-if)#ip add 10.2.25
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1
(23), with SwitchKleuter FastEthernet0/3 (99).
0
      ^
% Invalid input detected at '^' marker.

MLSkleuters(config-if)#ip add 10.2.250.6 255.255.255.252
MLSkleuters(config-if)#no shut
MLSkleuters(config-if)#exit
MLSkleuters(config)#

```

De switches een IP geven op de interfaces naar beide routers

```

MLSLager(config)#ip dhc
MLSLager(config)#ip dhcp exc
MLSLager(config)#ip dhcp excluded-address 10.2.10.2
MLSLager(config)#ip dhcp excluded-address 10.2.10.2
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1
(22), ip dhcp excluded-address 10.2.10.34
MLSLager(config)#ip dhcp excluded-address 10.2.10.34
MLSLager(config)#ip dhcp excluded-address 10.2.10.66
MLSLager(config)#ip dhcp excluded-address 10.2.10.66
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/3
(12), withip dhcp excluded-address 10.2.20.2
MLSLager(config)#ip dhcp excluded-address 10.2.21.19
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/5
(22), with SwitchITLager2 FastEthernet0/1 (99).
4
MLSLager(config)#ip dhcp excluded-address 10.2.21.194
MLSLager(config)#ip dhcp excluded-address 10.2.21.130
MLSLager(config)#ip dhcp excluded-address 10.2.22.2
MLSLager(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/4
(22), with SwitchITLager FastEthernet0/1 (99).
ip dhcp excluded-address 10.2.22.2
MLSLager(config)#ip dhcp excluded-address 10.2.30.2
MLSLager(config)#ip dhcp excluded-address 10.2.40.2
MLSLager(config)#ip dhcp excluded-address 10.2.40.2
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1
(22), ip dhcp excluded-address 10.2.41.2
MLSLager(config)#ip dhcp excluded-address 10.2.41.2
MLSLager(config)#ip dhcp excluded-address 10.2.42.2
MLSLager(config)#ip dhcp excluded-address 10.2.43.2
MLSLager(config)#ip dhcp excluded-address 10.2.44.2
MLSLager(config)#ip dhcp excluded-address 10.2.44.2
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/3
(12), withip dhcp excluded-address 10.2.99.2
MLSLager(config)#

```

Op iedere switch het eerste IP per subnet excluden uit de DHCP, om zo in de pkt file de ene verplichte statische IP te kunnen toekennen

```

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int gi0/0/0
R1(config-if)#ip add 10.2.250.1 255.255.255.252
R1(config-if)#standby 1 ip 10.2.250.254
% Warning: address is not within a subnet on this interface
R1(config-if)#standby 1 priority 110
R1(config-if)#standby 1 preempt
R1(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to down
R1(config-if)#int gi0/1/0
R1(config-if)#ip add 10.2.250.9 255.255.255.252
R1(config-if)#standby 2 ip 10.2.250.254
% Address 10.2.250.254 in group 1
R1(config-if)#standby 2 priority 110
R1(config-if)#standby 2 preempt
R1(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/1/0, changed state to down
R1(config-if)#int gi0/2/0
R1(config-if)#ip add 10.2.250.17 255.255.255.252
R1(config-if)#standby 3 ip 10.2.250.254
% Address 10.2.250.254 in group 1
R1(config-if)#standby 3 priority 110
R1(config-if)#standby 3 preempt
R1(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/2/0, changed state to down
R1(config-if)#ip route 10.2.0.0 255.255.0.0 10.2.250.2 100
R1(config)#ip route 10.2.0.0 255.255.0.0 10.2.250.10 150
R1(config)#ip route 10.2.0.0 255.255.0.0 10.2.250.18 200
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

```

IP's op de interface zetten alsook de priority aangepast zodat deze router als eerste gebruikt wordt. Daarnaast statische IP routes toegewezen richting de switches.

```

SKKlas3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SKKlas3(config)#int range Fa0/1 - 24
SKKlas3(config-if-range)#spanning-tree portfast

```

Zorgen dat elke poort op direct forwarding staat, en bescherming tegen het aansluiten van ongewenste switches.